

西方经济学

Part 2 Classical Theory

THE ECONOMY IN THE LONG RUN

Lecture 2A A Real Model of a Closed Economy

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西方经济学

(1) S3.1. ¹

(2) 其他资料: [GM]

- NATIONAL INCOME: HOW IT IS EARNED
- NATIONAL INCOME: HOW IT IS SPENT



¹M 指代马工程教材, S 指代课外阅读材料沈坤荣教程。

西方经济学

- (1) 掌握古典经济学中的实体经济模型。
- (2) 领会马工程教材精神。



Basic Assumptions in This Course

西方经济学

Assumption 1

For convenience, we assume $G^I = 0$, $NFP = 0$, $TR = 0$, and $INT^G = 0$.

It implies $G = G^C$, $\tilde{I} = I$, $GDP = GNP = Y$, $CA = NX$, $Y^{Dis} = Y - T$.

马克思主义理论研究和建设工程教材

西方经济学

(第二版)上册

《西方经济学》编写组



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Outline

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1 Classical Real Economy

- Production Function
- Markets for Factors
- Full Employment
- Distribution of National Income
- Loanable Funds Market
- Say's Law

2 马工程教材疑难重点



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Production Function: Homogeneity

Definition 1

A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $y = f(x_1, x_2, \dots, x_n)$, is said to be homogeneous of degree k if

$$f(t \cdot x_1, t \cdot x_2, \dots, t \cdot x_n) = t^k \cdot f(x_1, x_2, \dots, x_n). \quad (1)$$

Theorem 2

Suppose a differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $y = f(x_1, x_2, \dots, x_n)$, is homogeneous of degree k .

- i. The partial derivatives of f is homogeneous of degree $k-1$. That is, for every $j = 1, \dots, n$,

$$f_j(tx_1, \dots, tx_j, \dots, tx_n) = t^{k-1} \cdot f_j(x_1, \dots, x_j, \dots, x_n); \quad (2)$$

- ii. **Euler's theorem** holds:

$$kf(x_1, \dots, x_n) = x_1 f_1(x_1, \dots, x_n) + \dots + x_n f_n(x_1, \dots, x_n), \quad (3)$$

where f_j denotes the partial derivative of f with respect to the j th argument.



Neoclassical Production Function

Let $Y(t) = F[K(t), N(t), A(t)]$ denote a production function where Y is the flow of output produced at time t , K is the durable physical input or physical capital input at time t , N is the labor input at time t , and A is the level of knowledge or technology at time t . K and N are two rival inputs while A is a non-rival input.

Definition 3 (Neoclassical Production Function)

A production function $F: \mathbb{R}_+^3 \rightarrow \mathbb{R}_+$, $Y = F(K, N, A)$, twice differentiable in K and N , is *neoclassical* if it satisfies the following properties:

- i. Constant returns to scale or homogeneity of degree one in K and N .

$$F(\lambda K, \lambda N, A) = \lambda \cdot F(K, N, A), \quad \forall \lambda > 0.$$

- ii. Positive and diminishing returns to private inputs.

$$\begin{aligned} F_K(K, N, A) &> 0, \quad F_N(K, N, A) > 0, \\ F_{KK}(K, N, A) &< 0, \quad F_{NN}(K, N, A) < 0. \end{aligned}$$

- iii. Inada conditions.

$$\lim_{K \rightarrow 0^+} F_K = \lim_{N \rightarrow 0^+} F_N = +\infty, \quad \lim_{K \rightarrow \infty} F_K = \lim_{N \rightarrow \infty} F_N = 0.$$

Markets for Factors: Demands for Inputs

A firm has to make a decision of how much capital and labor it should employ. The objective of the firm is to maximize the economic profit.

$$\max_{K, N} P \times F(K, N, A) - W \times N - R \times K$$

The first-order conditions are

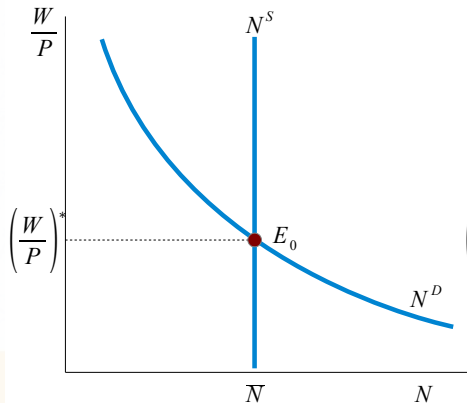
$$F_N(K, N, A) = W/P, \quad F_K(K, N, A) = R/P$$

which determine demands for capital and labor in the factor markets. The slopes of demand curves are negative since $F_{NN} < 0$ given K and $F_{KK} < 0$ given N .

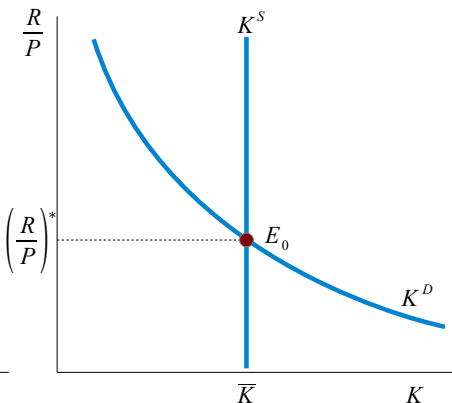


Markets for Factors: Determination of Factor Prices

The Labor Market



The Rental Market



Full Employment

Definition 4

The *full-employment level* of labor is an equilibrium level where the aggregate demand for labor is equal to the aggregate supply after the complete adjustment of nominal wage rate and the price of final goods. The output is called the *full-employment output*, *potential output* or *natural level of output* if labor is at full-employment level. The rate of unemployment corresponding to the full-employment level of labor is called the *natural rate of unemployment*.

The full employment of labor implies there is no *involuntary* unemployment. The definition of full employment can be generalized to other factor-markets.



Distribution of National Income

The maximized economic profit is

$$\text{Economic Profit} = P \times F(K, N, A) - P \times F_N \times N - P \times F_K \times K = 0$$

where the last equation uses homogeneity of degree one. Thus the economic profit is zero.

$$\text{Output } Y = MPN \times N + MPK \times K$$

= Labor Income

+ Capital Income + Depreciation of Capital

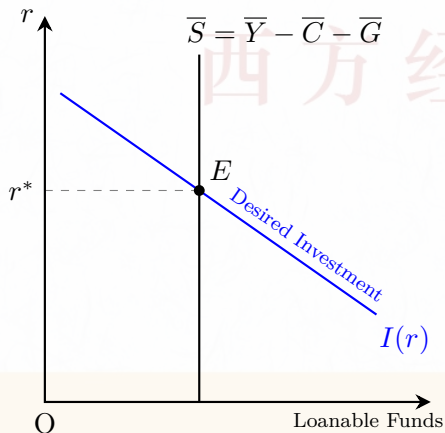
《资本论》第一卷，人民出版社，1975年，p57

第一章 商 品

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经常依靠自然力的帮助。因此，劳动并不是它所生产的使用价值即物质财富的唯一源泉。正象威廉·配第所说，劳动是财富之父，土地是财富之母。³¹

Loanable Funds Market: Equilibrium



The equilibrium in the factor market determines the output: $\bar{Y} = F(\bar{K}, \bar{N}, A)$. In equilibrium, $Y^S = Y^D$ implies

$$\bar{Y} = C(\bar{Y} - \bar{T}) + I(r) + \bar{G}$$

$$\bar{Y} - \bar{C} - \bar{G} = I(r)$$

$$(\bar{Y} - \bar{T} - \bar{C}) + (\bar{T} - \bar{G}) = I(r)$$

Private Saving + Public Saving = $I(r)$

National Saving $\bar{S} = I(r)$

It implies that the supply of loanable funds is equal to the demand for loanable funds.



Say's Law

Jean-Baptiste Say (1803, Book I, Chapter XV) introduced the idea in his work, *A Treatise on Political Economy*. (see [Wiki](#))

“It is worth while to remark, that a product is no sooner created, than it, from that instant, affords a market for other products to the full extent of its own value. When the producer has put the finishing hand to his product, he is most anxious to sell it immediately, lest its value should diminish in his hands. Nor is he less anxious to dispose of the money he may get for it; for the value of money is also perishable. But the only way of getting rid of money is in the purchase of some product or other. Thus, the mere circumstance of the creation of one product immediately opens a vent for other products.”

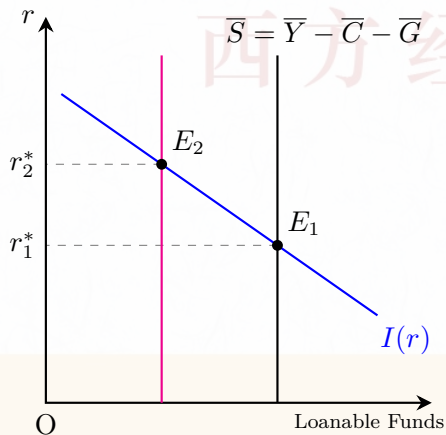
Proposition 5 (Say's Law)

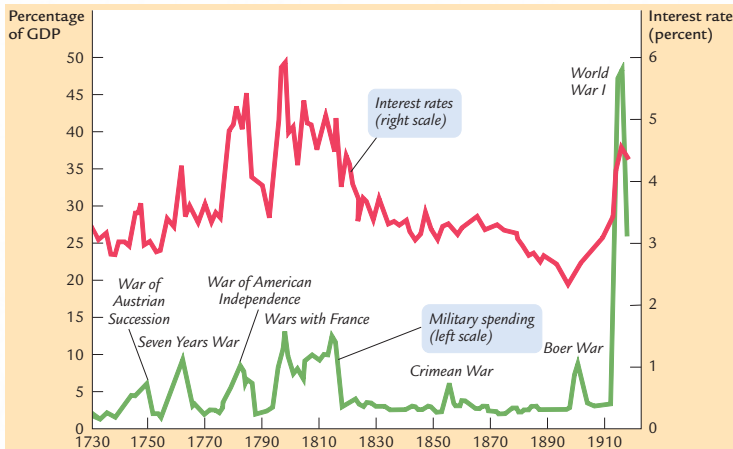
Supply creates its own demand.



Working with the Model

A Reduction in Saving





Military Spending and the Interest Rate in the United Kingdom from 1730 to 1919. Military spending is measured as a percentage of GDP.



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马克思对萨伊定律的批判

《马克思恩格斯全集》，第 26 卷（《资本论》第四卷“剩余价值理论”），第二册，人民出版社，1973 年，p.563

李嘉图的积累理论。对这个理论的批判

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考察资本的一般性质时，用不着对成为实际生产过程的一切前提的进一步的现实关系加以说明，就更加清楚地表现出来了。

[705] 大卫·李嘉图接受了庸俗的萨伊的(其实是属于詹姆斯·穆勒的)观点(我们谈这个微不足道的人物时，还要讲到这种观点)，认为生产过剩，至少市场商品普遍充斥是不可能的。这种观点是以产品同产品交换¹¹⁹这一论点为基础的，或者，正如穆勒所想象的那样，是以“卖者和买者之间的形而上学的平衡”¹²⁰为基础的，由此还进一步得出结论说，需求仅仅决定于生产本身，或者说，需求和供给完全一致。这种论点也采取李嘉图所特别喜爱的形式，即认为任何数额的资本在任何国家都能够生产地加以使用。

扩展阅读：

Baumol, William, J. 1999. "Retrospectives: Say's Law." *Journal of Economic Perspectives*, 13 (1): 195-204.



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疑难重点

西方经济学

- (1) 掌握生产函数的 Homogeneity 以及欧拉定理。
- (2) 什么是新古典生产函数？
- (3) 掌握要素市场需求曲线的推导。
- (4) 什么是充分就业？什么是潜在产出？什么是自然失业率？什么是非自愿失业？
- (5) 能够证明国民收入分配的公式。
- (6) 如何从可贷资金市场理解商品市场的均衡？
- (7) 商品市场上的总供给如何确定？总需求如何确定？
- (8) 什么是萨伊定律？



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马工程教材疑难重点

1 (E2, p.175)

根据马工程教材观点，应当如何评析西方经济学的失业理论？

2 (E2, p.311)

根据马工程教材观点，应当如何评析西方经济学的边际分配论？

3 (E2, p.223)

根据马工程教材观点，应当如何评析西方经济学的完全竞争假设？

4 (E2, p.188)

根据马工程教材观点，应当如何评析西方经济学的生产函数？

